

AI Lead Qualifier & Outreach Automation

An n8n automation that scores every inbound website lead with the BANT framework, replies within seconds, and follows up automatically on warm prospects who go quiet.

n8n

OpenAI

Google Sheets

Gmail

Slack

1 Executive Summary

Agencies, freelancers and consultants that receive leads through a web form typically lose time — and leads — to manual reading, scoring and replying. This workflow removes that bottleneck by automatically qualifying every lead the moment it arrives, replying with the right message within seconds, and following up on warm prospects who go quiet, all without a human touching the inbox.

2 Key Metrics

Trigger	Webhook (POST) from any website form or CRM
Qualification Engine	OpenAI GPT-4o-mini scoring 1–10 using BANT signals
Data Store	Google Sheets, used as a lightweight CRM
Outbound Channels	Gmail (personalised replies) + Slack (sales alerts)
Automation Cycle	Real-time intake plus a scheduled 9:00 AM daily follow-up job
Safeguards	Email validation, duplicate detection, private-URL blocking, prompt-injection guardrails
Lead Categories	Hot (8–10), Warm (5–7), Cold (1–4) — each with its own response and follow-up

1–10

BANT SCORE RANGE

9:00 AM

DAILY FOLLOW-UP

0

DUPLICATE OUTREACH

3 Business Value

- ✓ **Zero-delay response** — leads get a reply within moments of submitting the form, before interest cools.
- ✓ **Consistent qualification** — every lead is scored the same way, removing guesswork and rep-to-rep variance.
- ✓ **Sales team only sees what matters** — Slack alerts fire exclusively for Hot leads, cutting notification noise.
- ✓ **Nothing falls through the cracks** — the daily follow-up job automatically re-engages Hot/Warm leads gone quiet.

4 Architecture Overview

The workflow is split into two independent flows that share the same Google Sheet as their source of truth.

FLOW	TRIGGER	PURPOSE
New Lead Intake	Webhook, instant on form submit	Validate, deduplicate, enrich, AI-score and respond to a new lead
Daily Follow-up	Schedule, 9:00 AM daily	Re-engage Hot/Warm leads who have not replied within their follow-up window

5 How It Works

New Lead Intake

- 1 A webhook receives whatever the form or CRM sends — name, email, company, message, budget, website.
- 2 Fields are normalized, the email is format-checked, and private-network URLs are rejected before fetching.
- 3 The CRM is checked by email; an existing lead skips re-qualification and notifies Slack instead.
- 4 If a website was resolved, it's fetched and stripped to ~2,500 characters of context for the AI.
- 5 GPT-4o-mini scores the lead 1–10 on BANT signals and returns a category, reasoning and a reply opener.

Daily Follow-up

- 1 A 9:00 AM trigger pulls the full CRM and filters to leads whose follow-up window has passed.
- 2 Anyone already marked followed-up is excluded; if nothing qualifies, the run ends cleanly.
- 3 Gmail is searched for a reply from that lead before sending anything.
- 4 If they already replied, no follow-up is sent — the lead is simply marked handled.
- 5 Hot leads get a short reminder with the booking link; Warm leads get a friendly check-in.

6 Reliability & Security Design

- ◆ **Prompt-injection resistance** — the lead's message and scraped website text are labelled as untrusted evidence, never instructions.
- ◆ **SSRF protection** — localhost, loopback and private-network address ranges are rejected before any URL is fetched.
- ◆ **Fail-safe error handling** — a failed duplicate-check or website fetch degrades gracefully and notifies Slack.
- ◆ **Deterministic categorisation** — the Hot/Warm/Cold label is recalculated from the numeric score in code.
- ◆ **Idempotent CRM writes** — leads are upserted by email, preventing duplicate outreach even on repeated runs.
- ◆ **Reply-aware follow-ups** — Gmail is checked for an existing reply before any automated follow-up is sent.

7 Setup Checklist

- ☒ Create a Google Sheet Leads tab with the required headers (name, email, score, category, followed_up).
- ☒ Connect credentials: Google Sheets OAuth2, Gmail OAuth2, Slack OAuth2, OpenAI.
- ☒ Point every Google Sheets node at the same Sheet and tab, and select the Slack channel.
- ☒ Replace the placeholder booking link in the booking and follow-up emails.
- ☒ Test against the webhook's Test URL, then activate and switch to the Production URL.

8 Node-by-Node Map

Intake & Validation	Webhook - New Lead → Normalize Lead Data → IF Valid Email?
Dedup & Enrichment	Check Duplicate Lead → Fetch Company Website → Extract Website Text
AI Scoring & Routing	AI Qualify Lead (GPT-4o-mini) → Upsert CRM → Route by Category
Follow-up Job	Daily 9am Trigger → Filter Due Leads → Check for Reply → Send Follow-up Email

SUMMARY

Designed and built end-to-end in n8n, demonstrating webhook-driven intake, AI-based decisioning, CRM integration, and unattended scheduled processes.